

ESE5180: Lab 0 Zephyr

Team Member Name	Email Address
------------------	---------------

Deepa Lokesha	deepasr1@engineering.upenn.edu
---------------	--------------------------------

GitHub Repository URL: github.com/DeepaLokesha/lab0-zephyr-skeleton

(1.1) Create a video showing blinky on all 3x MCU boards.

drive.google.com/file/d/1MOml7tqNPhluztVFVqcjrf9GH49uKyGE/view?usp=drive_link

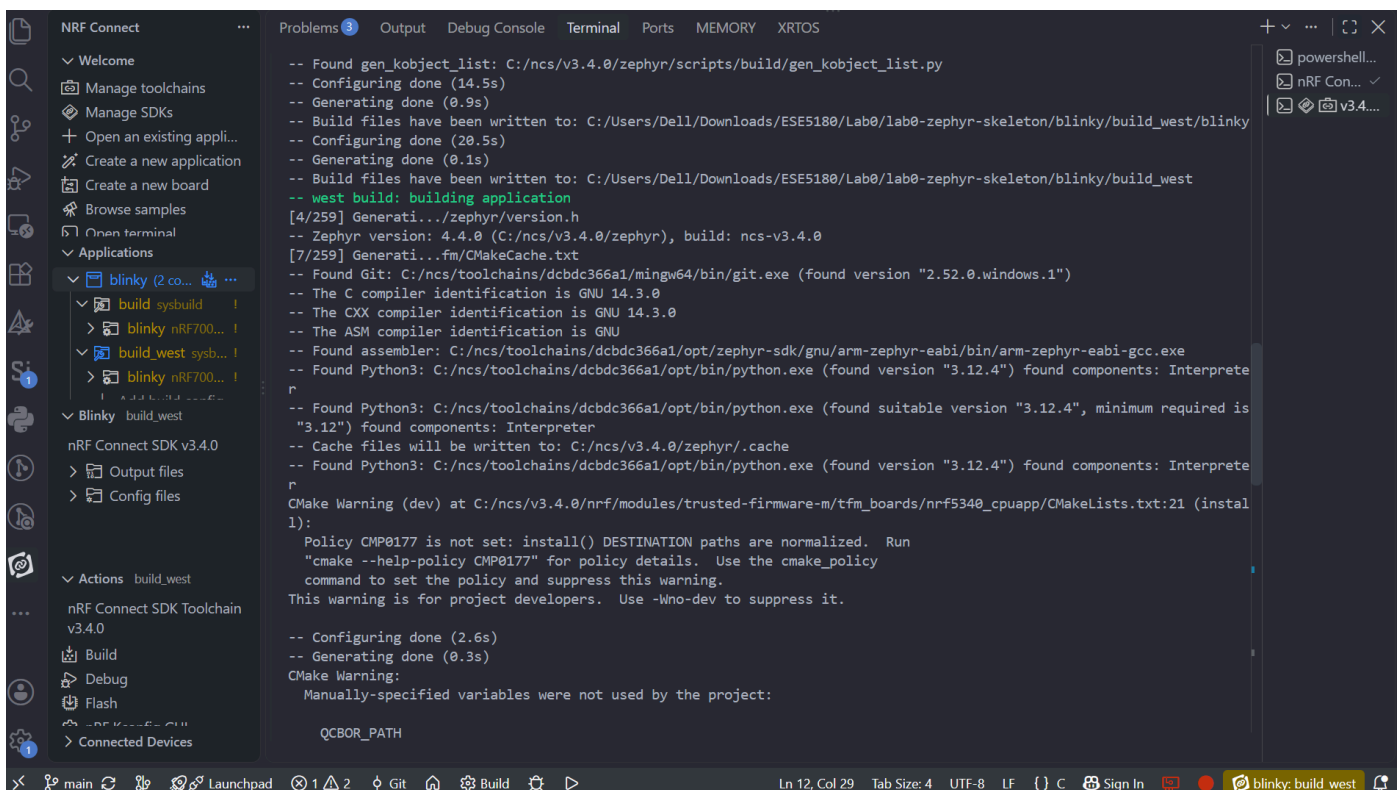
(2.1) Commit your Zephyr application to your GitHub repository

github.com/DeepaLokesha/lab0-zephyr-skeleton

(2.2) Create a video showing the change in blinky behavior on the nRF7002DK

drive.google.com/file/d/1olxMrAXM7og7ta9HKgVmGyFXPC0eONS7/view?usp=drive_link

(3.1) Build and Flash using only west commands and not the GUI. Show the terminal prints by embedding a screenshot in your README.md.



```
-- Found gen_kobject_list: C:/ncs/v3.4.0/zephyr/scripts/build/gen_kobject_list.py
-- Configuring done (14.5s)
-- Generating done (0.9s)
-- Build files have been written to: C:/Users/Dell/Downloads/ESE5180/Lab0/lab0-zephyr-skeleton/blinky/build_west/blinky
-- Configuring done (20.5s)
-- Generating done (0.1s)
-- Build files have been written to: C:/Users/Dell/Downloads/ESE5180/Lab0/lab0-zephyr-skeleton/blinky/build_west
-- west build: building application
[4/259] Generating.../zephyr/version.h
-- Zephyr version: 4.4.0 (C:/ncs/v3.4.0/zephyr), build: ncs-v3.4.0
[7/259] Generating...fm/CMakeCache.txt
-- Found Git: C:/ncs/toolchains/dcbdc366a1/mingw64/bin/git.exe (found version "2.52.0.windows.1")
-- The C compiler identification is GNU 14.3.0
-- The CXX compiler identification is GNU 14.3.0
-- The ASM compiler identification is GNU
-- Found assembler: C:/ncs/toolchains/dcbdc366a1/opt/zephyr-sdk/gnu/arm-zephyr-eabi/bin/arm-zephyr-eabi-gcc.exe
-- Found Python3: C:/ncs/toolchains/dcbdc366a1/opt/bin/python.exe (found version "3.12.4") found components: Interpreter
-- Found Python3: C:/ncs/toolchains/dcbdc366a1/opt/bin/python.exe (found suitable version "3.12.4", minimum required is "3.12") found components: Interpreter
-- Cache files will be written to: C:/ncs/v3.4.0/zephyr/.cache
-- Found Python3: C:/ncs/toolchains/dcbdc366a1/opt/bin/python.exe (found version "3.12.4") found components: Interpreter
CMake Warning (dev) at C:/ncs/v3.4.0/nrf/modules/trusted-firmware-m/tfm_boards/nrf5340_cpuapp/CMakeLists.txt:21 (install):
  Policy CMP0177 is not set: install() DESTINATION paths are normalized. Run
  "cmake --help-policy CMP0177" for policy details. Use the cmake_policy
  command to set the policy and suppress this warning.
This warning is for project developers. Use -Wno-dev to suppress it.

-- Configuring done (2.6s)
-- Generating done (0.3s)
CMake Warning:
  Manually-specified variables were not used by the project:

    QCBOR_PATH
```

File Edit Selection View Go Run ... < -> Untitled (Workspace) Update

Problems 3 Output Debug Console Terminal Ports MEMORY XRTOS

Policy CMP0177 is not set: install() DESTINATION paths are normalized. Run "cmake --help-policy CMP0177" for policy details. Use the cmake_policy command to set the policy and suppress this warning. This warning is for project developers. Use -Wno-dev to suppress it.

-- Configuring done (2.6s)
-- Generating done (0.3s)
CMake Warning:
Manually-specified variables were not used by the project:

QCBOR_PATH
TFM_CREATE_SIGNING_LAYOUT_OBJECTS
T_COSE_PATH

-- Build files have been written to: C:/Users/Dell/Downloads/ESE5180/Lab0/lab0-zephyr-skeleton/blink/ build_west/blink/ /tfm

[193/195] Linkin...ble bin\tfm_s.axf

Memory region	Used Size	Region Size	%age Used
FLASH:	81364 B	256 KB	31.04%
RAM:	45960 B	256 KB	17.53%

[195/195] Linkin...ibtfm_s_veneurs.a

[17/259] Perform...ll step for 'tfm'

-- Install configuration: "MinSizeRel"

----- Installing platform NS -----

[259/259] Linkin...zephyr\zephyr.elf

Memory region	Used Size	Region Size	%age Used
FLASH:	22576 B	192 KB	11.48%
RAM:	4600 B	192 KB	2.34%
IDT_LIST:	0 B	32 KB	0.00%

Generating files from C:/Users/Dell/Downloads/ESE5180/Lab0/lab0-zephyr-skeleton/blink/ build_west/blink/ zephyr/ zephyr.elf for board: nrf7002dk/nrf5340/cpuapp/ns

[8/8] Completed 'blink'

PS C:\Users\Dell\Downloads\ESE5180\west flash -d build_west --dev-id 001050794921 --erase

main Launchpad Git Build

Ln 12, Col 29 Tab Size: 4 UTF-8 LF {} C Sign In blinky: build_west

File Edit Selection View Go Run ... < -> Untitled (Workspace) Update

Problems 3 Output Debug Console Terminal Ports MEMORY XRTOS

[259/259] Linkin...zephyr\zephyr.elf

Memory region	Used Size	Region Size	%age Used
FLASH:	22576 B	192 KB	11.48%
RAM:	4600 B	192 KB	2.34%
IDT_LIST:	0 B	32 KB	0.00%

Generating files from C:/Users/Dell/Downloads/ESE5180/Lab0/lab0-zephyr-skeleton/blink/ build_west/blink/ zephyr/ zephyr.elf for board: nrf7002dk/nrf5340/cpuapp/ns

[8/8] Completed 'blink'

PS C:\Users\Dell\Downloads\ESE5180\west flash -d build_west --dev-id 001050794921 --erase

-- west flash: rebuilding

[0/16] Performing build step for 'tfm'ninja: no work to do.

[2/3] Performing install step for 'tfm'

-- Install configuration: "MinSizeRel"

----- Installing platform NS -----

[3/3] Completed 'tfm'

[3/3] Completed 'blink'

-- west flash: using runner nrfutil

-- runners.nrfutil: mass erase requested

-- runners.nrfutil: reset after flashing requested

-- runners.nrfutil: Flashing file: C:\Users\Dell\Downloads\ESE5180\Lab0\lab0-zephyr-skeleton\blink\ build_west\blink\ zephyr\ tfm_merged.hex

-- runners.nrfutil: Connecting to probe

-- runners.nrfutil: Erasing non-volatile memory (ERASEALL)

-- runners.nrfutil: Programming image

-- runners.nrfutil: Verifying image

-- runners.nrfutil: Reset

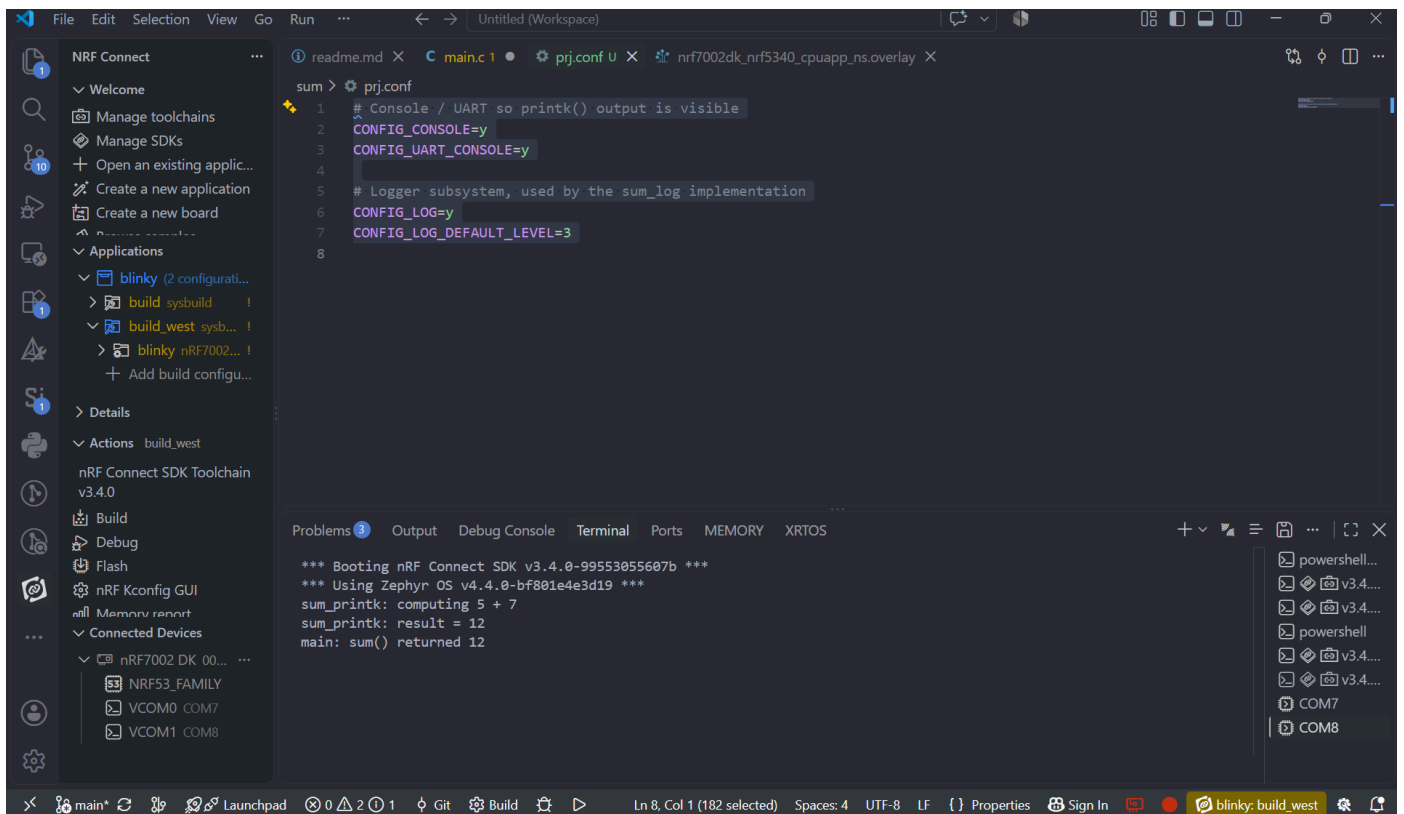
-- runners.nrfutil: Board(s) with serial number(s) 1050794921 flashed successfully.

PS C:\Users\Dell\Downloads\ESE5180\Lab0\lab0-zephyr-skeleton\blink>

main Launchpad Git Build

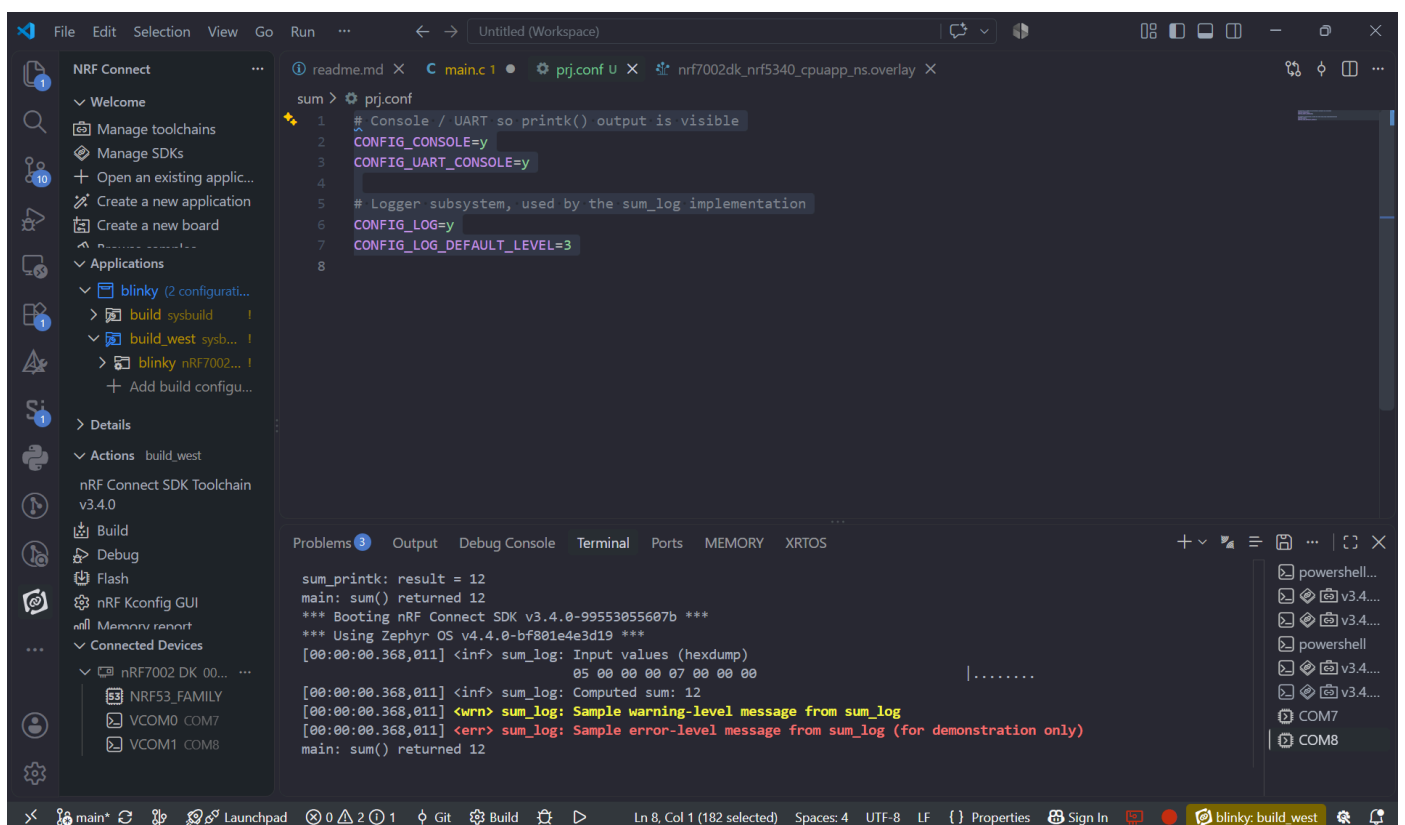
Ln 12, Col 29 Tab Size: 4 UTF-8 LF {} C Sign In blinky: build_west

(6.1) Take screenshots of console output for both builds:



```
sum > prj.conf
1 # Console / UART so printk() output is visible
2 CONFIG_CONSOLE=y
3 CONFIG_UART_CONSOLE=y
4
5 # Logger subsystem, used by the sum_log implementation
6 CONFIG_LOG=y
7 CONFIG_LOG_DEFAULT_LEVEL=3
8

*** Booting nRF Connect SDK v3.4.0-99553055607b ***
*** Using Zephyr OS v4.4.0-bf801e4e3d19 ***
sum_printk: computing 5 + 7
sum_printk: result = 12
main: sum() returned 12
```



```
sum > prj.conf
1 # Console / UART so printk() output is visible
2 CONFIG_CONSOLE=y
3 CONFIG_UART_CONSOLE=y
4
5 # Logger subsystem, used by the sum_log implementation
6 CONFIG_LOG=y
7 CONFIG_LOG_DEFAULT_LEVEL=3
8

sum_printk: result = 12
main: sum() returned 12
*** Booting nRF Connect SDK v3.4.0-99553055607b ***
*** Using Zephyr OS v4.4.0-bf801e4e3d19 ***
[00:00:00.368,011] <inf> sum_log: Input values (hexdump)
                                05 00 00 00 07 00 00 00
[00:00:00.368,011] <inf> sum_log: Computed sum: 12
[00:00:00.368,011] <wrn> sum_log: Sample warning-level message from sum_log
[00:00:00.368,011] <err> sum_log: Sample error-level message from sum_log (for demonstration only)
main: sum() returned 12
```

(6.2) Make a short video showing hexdump, log, and printk.

drive.google.com/file/d/1dpdtQi58VGnqcgS_g0-UkT-9CTGCYmNv/view?usp=drive_link